

# FRANCESCO GANIS

## PhD, Sound & Music Engineer

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francescoganis.github.io

Francesco Ganis

Copenhagen, DK



With a strong background in audio digital signal processing (DSP), prototyping, and the evaluation of audio solutions, I bring experience in developing user-centered audio technologies and vibrotactile feedback systems. I am a passionate audio engineer and researcher, with deep love for music and sound. I am eager to contribute, while continuing to expand my knowledge and skills. You can find my portfolio at [francescoganis.github.io](https://francescoganis.github.io).

## EDUCATION

### PhD, Music Training Tools for Children with Hearing Impairment

#### Aalborg University

Sep. 2022 – Mar. 2026

Copenhagen, DK

Skills learned Research Prototyping Evaluating

- Designing, developing, and evaluating musical training solutions for children with hearing loss, including:
  - vibrotactile feedback prototypes
  - video game with vibrotactile feedback for timbre recognition
  - scalable training tool for sound localization in children
  - training tool for emotion recognition in children

### Master's Degree, Sound & Music Computing

#### Aalborg University

Sep. 2020 – Jun. 2022

Copenhagen, DK

Skills learned DSP Prototyping Evaluating

- Thesis: "Vibrotactile Feedback for Musical Training of Cochlear Implants Users"
- Completed courses in Sound Processing, Embodied Interaction, Music Perception and Cognition, New Interfaces for Musical Expression

### Bachelor's Degree, Electronic Music & Sound Engineering

#### "C. Pollini" Conservatory

Sep. 2016 – Apr. 2020

Padova, IT

Skills learned Sound Engineering MAX/MSP Sound Design

- Graduated with honors
- Thesis: "Continuous Sonification of Autonomous Driving Vehicles with Purposes of Passengers Information and Trust Augmentation"

## EXPERIENCE

### Internship, Technology Evaluation Engineer

#### Bang & Olufsen

Apr. 2026 – Ongoing

Lyngby, DK

Skills learned Product Evaluation Perceptual Testing  
Technical Documentation

## STRENGTHS

### Technical Skills

MATLAB Prototyping DSP  
Product Evaluation Audio Testing  
C Arduino Git Unity C#  
Embedded Programming ESP32  
LaTeX Lasercutting MAX/MSP  
Pro Tools Pure Data Reaper  
Sound Engineering 3D Printing

### Soft Skills

Analytical Thinking Collaboration  
Detail-oriented Problem-solving  
Scientific Dissemination Supervision  
Teamwork Technical Documentation

## OTHER EXPERTISE

Photography Playing Drums

## LANGUAGES

Italian ● ● ● ● ●  
Mother tongue

English ● ● ● ● ●  
Full professional proficiency

## PUBLICATIONS

### Journal Articles

- F. Ganis et al., "The role of haptics in training and games for hearing-impaired individuals: A systematic review," *Multi-modal Technologies and Interaction*, vol. 8, no. 1, p. 1, 2023.

- Development of experimental paradigm for the perceptual evaluation of high-end audio products

## Teaching Assistant, Sound & Music Computing

### Aalborg University

📅 Sep. 2022 – Aug. 2025

📍 Copenhagen, DK

Skills learned Mentoring Communication Supervision

- Assisting students in the *Sound Processing* course
- Assisting students in the *Physical Models for Sound Synthesis* course
- Supervision of student projects in Medialogy and Sound & Music Computing programs

## Student Helper, Sound Synthesis and Design

### Aalborg University

📅 Oct. 2021 – Apr. 2022

📍 Copenhagen, DK

Skills learned Audio Programming DSP Web Development

- Created sound examples in MATLAB for the book *Serafin, S., Buxton, B., Gaver, B., & Bly, S. (2022). Auditory interfaces. Focal Press*
- Developed and maintained websites

## Internship, Prototype Development

### Oticon Medical

📅 Oct. 2021 – Jan. 2022

📍 Copenhagen, DK

Skills learned Product Development Prototyping Evaluation

- Development of smartphone cover that provides high-quality vibrotactile feedback to support music listening for individuals with hearing impairment

## ICT Technician

### GAMA S.r.l.

📅 Dec. 2019 – Jan. 2020

📍 Tavagnacco, IT

Skills learned Technical Support Hardware Maintenance

- Audio and conferencing systems' installation and maintenance
- Assembling and repairing PCs, workstations and small servers

## Sound Engineer, Studio and Live Sound Engineering

### Various

📅 2017 – 2018

📍 Veneto Region, IT

Skills learned Sound Engineering Live Electronics

- Centro D'Arte & C. Pollini Conservatory - *Live Electronics, Sound Enigineering for 11 Channels Sound System*
- RmS Recording Studio - *Boom Operator, Editing Technician*
- Bluetrain's Studio - *FoH Sound Engineer, Live Stage Technician, Live Video Director*

## Internship, Industrial Prototyping

### Danieli Automation S.p.A.

📅 Jun. 2013 – Jul. 2013

📍 Buttrio, IT

Skills learned Prototyping Industrial Design

- Design and creation of a working prototype of a measurement system for foundry equipment

## 👥 Conference Proceedings

- F. Ganis et al., "Vibrotactile teddy bear: Enhancing musical experiences for children with cochlear nerve deficiency through a vibrotactile soft toy," *SMC Proceedings*, 2025, pp. 115–120.
- F. Ganis et al., "Vibrotactile memory: A case study of timbre perception training in children with cochlear implants using a video game," *SMC Proceedings*, 2024, pp. 134–141.
- F. Ganis et al., "Tickle tuner-haptic smart-phone cover for cochlear implant users' musical training," in *HAID*, Springer Nature, 2022, p. 14.
- F. Ganis et al., "Real-time timbre transfer and sound synthesis using ddsdp," *SMC Proceedings*, 2021, pp. 175–182.

## 🎵 Albums

- M. Finco et al., *Six flute sonatas op.1*, album, EAN Code: 7.93588412371, Japan, 2018.

## FURTHER TRAINING

### Masterclass, Live Electronics

#### Chigiana Academy

📅 Jul. 2019

📍 Siena, IT

### Masterclass, Symphonic Orchestra Recording

#### G. A. Fano Institute

📅 Jul. 2018

📍 Spilimbergo, IT

### Diploma, Music, Theory and Solfège

#### G. Tartini Conservatory

📅 Sep. 2009

📍 Trieste, IT

## REFERENCES

### Prof. Stefania Serafin

@ Technical University of Denmark, Konges Lyngby - DK

✉ stefse@dtu.dk

### Prof. Lone Percy-Smith

@ Righospitalet, Copenhagen - DK

✉ lone.percy-smith@regionh.dk

### Prof. Nicola Bernardini

@ Conservatorio di Latina, Latina - IT

✉ nicb@sme-ccppd.org